Multi-orbital dynamical vertex approximation (DΓA)

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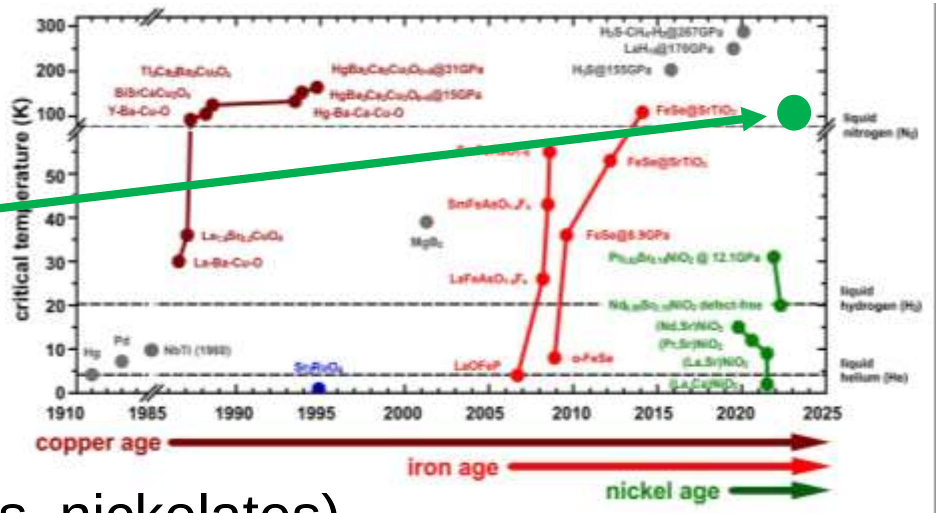
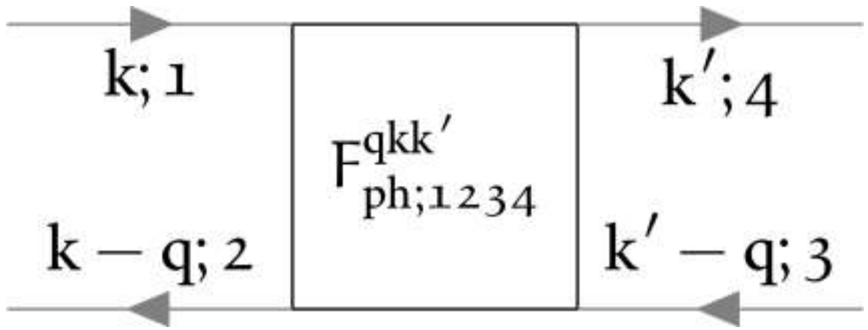
$La_3Ni_2O_7$
@ 14 GPa



Motivation

- Unconventional superconductivity (e.g. cuprates, nickelates) requires treatment of non-local correlations beyond DMFT → DΓA
- Currently, DΓA & superconductivity calculations only implemented for single bands [1, 2]
- Many correlated materials (e.g. $La_3Ni_2O_7$) need a multi-orbital description [3]
- Calculations down to superconducting transition temperatures → vertex asymptotics [4]

[1] P. Worm, 10.5281/zenodo.10406493
[2] A. Galler et al., 10.1016/j.cpc.2019.07.012
[3] H.Sun et al., 10.1038/s41586-023-06408-7
[4] M. Kitatani et al., 10.1088/2515-7639/ac7e6d



Implementation details [5]

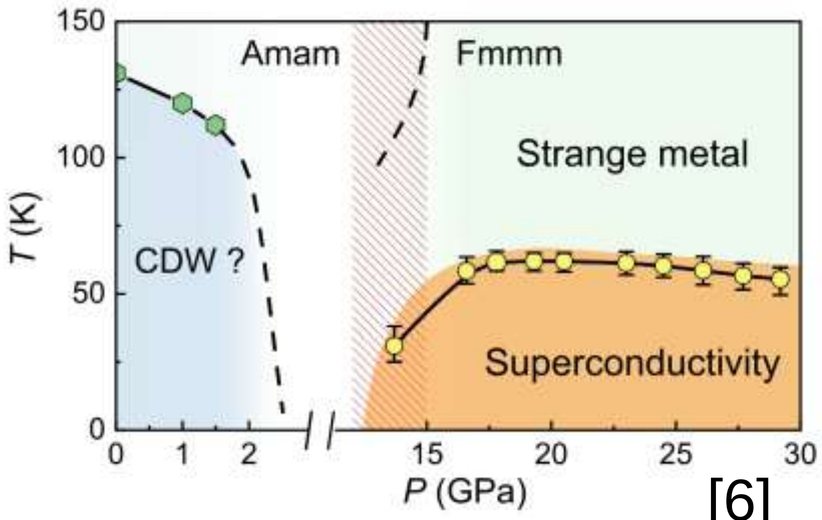
- From single- to multi-orbital:
  - Green's functions & self-energies gain two orbital indices: $G^{k} \rightarrow G^{k}_{12}$
  - Two-particle vertex functions gain four orbital indices: $F^{qkk'} \rightarrow F^{qkk'}_{1234}$
  - Memory requirements ↑ and computation time ↑
- Multiplications include orbital contractions, e.g. $C^{qkk'}_{1234} = A^{qkk_1}_{12ab} B^{qk_1k'}_{ba34}$
- Parallelized code that takes advantage of system symmetries (e.g. full → irreducible Brillouin zone, frequency symmetries, …)

[5] https://github.com/Julpe/moLDGA

Benchmarks

- Agreement with single-band code up to numerical precision [1] ✓
- Correct decoupling for disconnected two-band tests ✓
- Performance: ~12 h → ~30 min for low-temperature data compared to [1]

Outlook



[6]

- Testing limits of the code
- Benchmarks against multi-orbital calculations performed by [2] ($SrVO_3$)
- Calculation of realistic materials (especially $La_3Ni_2O_7$)

[1] P. Worm, 10.5281/zenodo.10406493
[2] A. Galler et al., 10.1016/j.cpc.2019.07.012
[6] Y. Zhang et al., 10.1038/s41567-024-02515-y

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Thank you for your attention!
Any questions?